

РЕФЕРАТ

Название программы для ЭВМ

Tokamak Analysis — ИИ-система для анализа экспериментальных данных токамаков

Правообладатель

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Авторы

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Год создания

2026

Язык программирования

Python 3.11, TypeScript 5.6

Объём программы

~24 000 строк исходного кода

Операционная система

Linux (Ubuntu 22.04/24.04), кроссплатформенная (Docker)

Объём документации

4 документа (README.md, ARCHITECTURE.md, CONTRIBUTING.md, CLAUDE.md)

Аннотация

Программа «Tokamak Analysis» представляет собой веб-платформу на основе технологий машинного обучения, предназначенную для анализа экспериментальных данных установок управляемого термоядерного синтеза (токамаков). Платформа реализована в виде микросервисной архитектуры, состоящей из модульного монолитного бэкенда (FastAPI, Python 3.11), выделенного ML-сервиса (PyTorch 2.x, scikit-learn), веб-интерфейса (React 18, TypeScript 5.6) и инфраструктурных сервисов (PostgreSQL 16, Redis 7, Celery, MinIO,

Nginx, Prometheus, Grafana), развёрнутых в контейнерах Docker Compose (12 сервисов, 4 изолированные сети).

Основные функции программы:

1. Загрузка и предобработка экспериментальных данных из международного архива FAIR-MAST (11 573 экспериментальных «выстрела» токамака MAST, Великобритания) через протокол S3/Zarr.
2. Интерактивная визуализация временных рядов параметров плазмы (ток плазмы, электронная плотность, температура, коэффициент запаса устойчивости q_{95} и др.) с использованием библиотеки Plotly.js.
3. Классификация режимов плазмы (стабильный/нестабильный) с использованием трёх архитектур машинного обучения:
 - Random Forest (scikit-learn) — базовая модель, точность 97.3%, $F1 = 0.973$, $AUC-ROC = 0.9677$;
 - Bidirectional LSTM с механизмом multi-head attention (PyTorch) — лучшая модель, точность 97.3%, $F1 = 0.973$, $AUC-ROC = 0.9938$;
 - Autoregressive Transformer с позиционным кодированием RoPE (PyTorch) — точность 97.3%, $F1 = 0.973$, $AUC-ROC = 0.9415$.Дополнительно реализованы модели прогнозирования временных рядов (forecasting): LSTM Forecaster (NRMSE = 0.143) и Transformer Forecaster (NRMSE = 0.196) для бенчмарка TokaMark.
4. Предсказание срывов плазмы с заблаговременностью не менее 30 мс до начала токового срыва (медианное время предупреждения — 47 мс для модели bi-LSTM) и формирование рекомендаций по стабилизации (снижение тока, газонапуск, коррекция формы плазмы).
5. Асинхронное обучение моделей с возможностью настройки гиперпараметров, версионирование моделей в объектном хранилище MinIO (S3-совместимое).
6. Пакетная обработка нескольких экспериментов.
7. Экспорт результатов в форматах CSV и JSON.
8. Управление пользователями с ролевой моделью доступа (исследователь, инженер, администратор), аутентификация JWT (HS256, bcrypt cost 12), соответствие 152-ФЗ «О персональных данных».
9. Мониторинг производительности (Prometheus + Grafana) с предустановленными дашбордами.

Технический стек: Python 3.11, PyTorch 2.x, scikit-learn, FastAPI, SQLAlchemy (async), PostgreSQL 16, Redis 7, Celery, React 18, TypeScript 5.6, Vite, Plotly.js, Zustand, Docker Compose, Nginx, MinIO, Prometheus, Grafana. Все модели реализуют единый протокол ModelInterface (fit, predict, predict_proba, save, load, metadata). Время вывода (inference) для всех моделей не превышает 50 мс (P99), что удовлетворяет требованиям к системам реального времени.

Экспериментальная валидация проведена на выборке из 500 выстрелов MAST (20 параметров плазмы, 200 временных шагов). Темпоральная валидация (обучение на ранних, тест на поздних кампаниях) показала снижение AUC-ROC на 3–27%, подтверждая необходимость хронологического разделения данных. Модель LSTM Forecaster показала NRMSE = 0.143 на задачах TokaMark, превосходя baseline Group 1 (NRMSE = 0.163).

Программа является свободным программным обеспечением (лицензия MIT).

Область применения: научно-исследовательские организации, занимающиеся исследованиями управляемого термоядерного синтеза (Росатом, ИТЭР, UKAEA, EUROfusion), а также образовательные учреждения для подготовки кадров в области ИИ и физики плазмы.

Ключевые слова

Токамак, плазма, срыв, машинное обучение, нейронная сеть, LSTM, Transformer, Random Forest, термоядерный синтез, FAIR-MAST, предсказание срывов, классификация, визуализация

Депонируемые материалы

Программа для ЭВМ «Tokamak Analysis»

ДЕПОНИРУЕМЫЕ МАТЕРИАЛЫ

Программа для ЭВМ: Tokamak Analysis — ИИ-система для анализа
экспериментальных данных токамаков

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Дата создания: 2026

Дата формирования листинга: 05.05.2026

Языки программирования: Python 3.11, TypeScript 5.6

Объем: ~14 000 строк исходного кода

ФАЙЛ: backend/app/main.py

Описание: Точка входа серверного приложения (FastAPI), middleware, маршруты

```
from __future__ import annotations

import time
from contextlib import asynccontextmanager

import structlog
from fastapi import FastAPI, Request
from fastapi.exceptions import RequestValidationError
from fastapi.middleware.cors import CORSMiddleware
from fastapi.responses import JSONResponse, RedirectResponse
from slowapi import Limiter, _rate_limit_exceeded_handler
from slowapi.errors import RateLimitExceeded
from slowapi.util import get_remote_address
from starlette.middleware.base import BaseHTTPMiddleware, RequestResponseEndpoint
from starlette.responses import Response

from app.api import admin, auth, experiments, legal, models, predictions
from app.config import settings
from app.core.headers import SecurityHeadersMiddleware
from app.core.logging_config import setup_logging
from app.core.metrics import instrumentator
from app.services.ml_client import MLServiceClient

logger = structlog.get_logger()

class RequestLoggingMiddleware(BaseHTTPMiddleware):
    """Log method, path, status code and duration for every request."""

    async def dispatch(self, request: Request, call_next: RequestResponseEndpoint)
-> Response:
        start = time.perf_counter()
        response = await call_next(request)
        duration_ms = round((time.perf_counter() - start) * 1000, 2)
```

```

        await logger.ainfo(
            "request",
            method=request.method,
            path=request.url.path,
            status=response.status_code,
            duration_ms=duration_ms,
        )
    return response

```

```

@asynccontextmanager
async def lifespan(app: FastAPI):
    # Startup
    setup_logging()
    from app.database import engine
    yield
    # Shutdown
    await engine.dispose()

```

```

limiter = Limiter(key_func=get_remote_address,
default_limits=[f"{settings.rate_limit_per_minute}/minute"])

```

```

app = FastAPI(
    title="Tokamak Analysis API",
    description="ИИ-система для анализа экспериментальных данных токамаков",
    version="1.0.0",
    lifespan=lifespan,
)

```

```

app.state.limiter = limiter
app.add_exception_handler(RateLimitExceeded, _rate_limit_exceeded_handler)

```

```

@app.exception_handler(RequestValidationError)
async def validation_error_handler(request: Request, exc: RequestValidationError)
-> JSONResponse:
    return JSONResponse(
        status_code=422,
        content={"detail": exc.errors()},
    )

```

```

# Middleware (order matters: outermost middleware is added last)
app.add_middleware(RequestLoggingMiddleware)
app.add_middleware(SecurityHeadersMiddleware)

```

```

# CORS
origins = [o.strip() for o in settings.backend_cors_origins.split(",")]
app.add_middleware(
    CORSMiddleware,
    allow_origins=origins,
    allow_credentials=True,
    allow_methods=["*"],
    allow_headers=["*"],
)

```

```

# Routers

```

```

app.include_router(auth.router)
app.include_router(experiments.router)
app.include_router(predictions.router)
app.include_router(models.router)
app.include_router(admin.router)
app.include_router(legal.router)

# Prometheus metrics
instrumentator.instrument(app).expose(app)

@app.get("/api/v1/health")
async def health():
    ml_client = MLServiceClient()
    ml_status = await ml_client.health()
    return {
        "status": "healthy",
        "version": "1.0.0",
        "ml_service": "connected" if ml_status else "unavailable",
    }

@app.get("/api/docs")
async def api_docs_redirect():
    return RedirectResponse(url="/docs")

@app.get("/health")
async def health_simple():
    return {"status": "ok"}

```

```

=====
ФАЙЛ: backend/app/api/auth.py
Описание: API аутентификации (регистрация, JWT, 152-ФЗ)
=====

```

```

from __future__ import annotations

from datetime import datetime, timezone

from fastapi import APIRouter, Depends, HTTPException, Request, status
from sqlalchemy import delete, select
from sqlalchemy.ext.asyncio import AsyncSession
from app.core.audit import log_action
from app.core.security import (
    create_access_token,
    create_refresh_token,
    decode_token,
    get_current_user,
    hash_password,
    verify_password,
)
from app.database import get_db
from app.models.audit_log import AuditLog
from app.models.experiment import Experiment, TimeSeriesData
from app.models.ml_model import ModelRun
from app.models.prediction import Prediction

```

```

from app.models.user import User
from app.schemas.auth import (
    TokenRefresh,
    TokenResponse,
    UserCreate,
    UserLogin,
    UserResponse,
)

router = APIRouter(prefix="/api/v1/auth", tags=["auth"])

@router.post("/register", response_model=UserResponse,
status_code=status.HTTP_201_CREATED)
async def register(
    data: UserCreate,
    request: Request,
    db: AsyncSession = Depends(get_db),
):
    if not data.consent_given:
        raise HTTPException(
            status_code=status.HTTP_400_BAD_REQUEST,
            detail="Необходимо согласие на обработку персональных данных (152-ФЗ)",
        )

    existing = await db.execute(select(User).where(User.email == data.email))
    if existing.scalar_one_or_none():
        raise HTTPException(status_code=status.HTTP_409_CONFLICT, detail="Email
already registered")

    user = User(
        email=data.email,
        password_hash=hash_password(data.password),
        full_name=data.full_name,
        role=data.role,
        consent_given_at=datetime.now(timezone.utc),
    )
    db.add(user)
    await db.flush()

    await log_action(db, "register", "user", user_id=user.id, request=request)
    return user

@router.post("/login", response_model=TokenResponse)
async def login(
    data: UserLogin,
    request: Request,
    db: AsyncSession = Depends(get_db),
):
    result = await db.execute(select(User).where(User.email == data.email))
    user = result.scalar_one_or_none()

    if not user or not verify_password(data.password, user.password_hash):
        await log_action(
            db, "login_failed", "auth",
            details={"email": data.email},
            request=request,

```

```

        )
        await db.commit()
        raise HTTPException(status_code=status.HTTP_401_UNAUTHORIZED,
detail="Invalid credentials")

    if not user.is_active:
        raise HTTPException(status_code=status.HTTP_403_FORBIDDEN, detail="Account
disabled")

    await log_action(db, "login", "auth", user_id=user.id, request=request)

    return TokenResponse(
        access_token=create_access_token(user.id, user.role.value),
        refresh_token=create_refresh_token(user.id),
    )

@router.post("/refresh", response_model=TokenResponse)
async def refresh_token(
    data: TokenRefresh,
    db: AsyncSession = Depends(get_db),
):
    payload = decode_token(data.refresh_token)
    if payload.get("type") != "refresh":
        raise HTTPException(status_code=status.HTTP_401_UNAUTHORIZED,
detail="Invalid token type")

    user_id = int(payload["sub"])
    result = await db.execute(select(User).where(User.id == user_id,
User.is_active.is_(True)))
    user = result.scalar_one_or_none()
    if not user:
        raise HTTPException(status_code=status.HTTP_401_UNAUTHORIZED, detail="User
not found")

    return TokenResponse(
        access_token=create_access_token(user.id, user.role.value),
        refresh_token=create_refresh_token(user.id),
    )

@router.get("/me", response_model=UserResponse)
async def get_me(current_user: User = Depends(get_current_user)):
    return current_user

@router.delete("/account", status_code=status.HTTP_200_OK)
async def delete_account(
    request: Request,
    current_user: User = Depends(get_current_user),
    db: AsyncSession = Depends(get_db),
):
    """Delete current user account and all associated data (152-Ø3, ст. 21)."""
    user_id = current_user.id

    # Delete predictions linked to user's experiments
    user_experiment_ids = select(Experiment.id).where(Experiment.user_id ==
user_id)

```



```

        await db.execute(
            delete(Prediction).where(Prediction.experiment_id.in_(user_experiment_ids))
        )

        # Delete timeseries data linked to user's experiments
        await db.execute(
            delete(TimeSeriesData).where(TimeSeriesData.experiment_id.in_(user_experiment_ids))
        )

        # Delete user's experiments
        await db.execute(delete(Experiment).where(Experiment.user_id == user_id))

        # Delete user's model runs
        await db.execute(delete(ModelRun).where(ModelRun.user_id == user_id))

        # Delete user's audit logs
        await db.execute(delete(AuditLog).where(AuditLog.user_id == user_id))

        # Soft-delete the user account
        current_user.is_active = False
        current_user.deleted_at = datetime.now(timezone.utc)

        await log_action(
            db, "account_deleted", "user",
            user_id=user_id,
            details={"email": current_user.email},
            request=request,
        )
        await db.commit()

        return {"detail": "Аккаунт и все связанные данные удалены"}

@router.get("/export-data")
async def export_data(
    current_user: User = Depends(get_current_user),
    db: AsyncSession = Depends(get_db),
):
    """Export all user personal data (152-ФЗ, ст. 14 / right to data
    portability)."""
    # Profile
    profile = {
        "id": current_user.id,
        "email": current_user.email,
        "full_name": current_user.full_name,
        "role": current_user.role.value,
        "is_active": current_user.is_active,
        "created_at": current_user.created_at.isoformat() if
current_user.created_at else None,
        "consent_given_at": current_user.consent_given_at.isoformat() if
current_user.consent_given_at else None,
    }

    # Experiments
    result = await db.execute(
        select(Experiment).where(Experiment.user_id == current_user.id)
    )

```

```

experiments = result.scalars().all()
experiments_data = [
    {
        "id": exp.id,
        "shot_id": exp.shot_id,
        "source": exp.source.value,
        "status": exp.status.value,
        "metadata": exp.metadata_json,
        "loaded_at": exp.loaded_at.isoformat() if exp.loaded_at else None,
    }
    for exp in experiments
]

# Predictions linked to user's experiments
experiment_ids = [exp.id for exp in experiments]
predictions_data = []
if experiment_ids:
    result = await db.execute(
        select(Prediction).where(Prediction.experiment_id.in_(experiment_ids))
    )
    predictions = result.scalars().all()
    predictions_data = [
        {
            "id": pred.id,
            "experiment_id": pred.experiment_id,
            "model_id": pred.model_id,
            "result": pred.result_json,
            "probability": pred.probability,
            "created_at": pred.created_at.isoformat() if pred.created_at else
None,
        }
        for pred in predictions
    ]

# Model runs
result = await db.execute(
    select(ModelRun).where(ModelRun.user_id == current_user.id)
)
model_runs = result.scalars().all()
model_runs_data = [
    {
        "id": run.id,
        "model_id": run.model_id,
        "status": run.status.value,
        "hyperparams": run.hyperparams_json,
        "metrics": run.metrics_json,
        "progress": run.progress,
        "started_at": run.started_at.isoformat() if run.started_at else None,
        "finished_at": run.finished_at.isoformat() if run.finished_at else
None,
        "created_at": run.created_at.isoformat() if run.created_at else None,
    }
    for run in model_runs
]

return {
    "profile": profile,
    "experiments": experiments_data,

```

```
        "predictions": predictions_data,
        "model_runs": model_runs_data,
    }
```

```
=====
ФАЙЛ: backend/app/core/security.py
```

```
Описание: Аутентификация JWT, хеширование паролей (bcrypt), RBAC
=====
```

```
from __future__ import annotations
```

```
from datetime import datetime, timedelta, timezone
```

```
from fastapi import Depends, HTTPException, Request, status
```

```
from fastapi.security import HTTPAuthorizationCredentials, HTTPBearer
```

```
from jose import JWTError, jwt
```

```
from passlib.context import CryptContext
```

```
from sqlalchemy import select
```

```
from sqlalchemy.ext.asyncio import AsyncSession
```

```
from app.config import settings
```

```
from app.database import get_db
```

```
from app.models.user import User, UserRole
```

```
pwd_context = CryptContext(schemes=["bcrypt"], deprecated="auto",
```

```
bcrypt__rounds=12)
```

```
security_scheme = HTTPBearer()
```

```
def hash_password(password: str) -> str:
```

```
    return pwd_context.hash(password)
```

```
def verify_password(plain_password: str, hashed_password: str) -> bool:
```

```
    return pwd_context.verify(plain_password, hashed_password)
```

```
def create_access_token(user_id: int, role: str) -> str:
```

```
    expire = datetime.now(timezone.utc) +
```

```
    timedelta(minutes=settings.jwt_access_token_expire_minutes)
```

```
    payload = {"sub": str(user_id), "role": role, "type": "access", "exp": expire}
```

```
    return jwt.encode(payload, settings.jwt_secret_key,
```

```
    algorithm=settings.jwt_algorithm)
```

```
def create_refresh_token(user_id: int) -> str:
```

```
    expire = datetime.now(timezone.utc) +
```

```
    timedelta(days=settings.jwt_refresh_token_expire_days)
```

```
    payload = {"sub": str(user_id), "type": "refresh", "exp": expire}
```

```
    return jwt.encode(payload, settings.jwt_secret_key,
```

```
    algorithm=settings.jwt_algorithm)
```

```
def decode_token(token: str) -> dict:
```

```
    try:
```

```
        return jwt.decode(token, settings.jwt_secret_key,
```

```
        algorithms=[settings.jwt_algorithm])
```

```

        except JWTError as e:
            raise HTTPException(status_code=status.HTTP_401_UNAUTHORIZED,
                                detail="Invalid token") from e

    async def get_current_user(
        credentials: HTTPAuthorizationCredentials = Depends(get_security_scheme),
        db: AsyncSession = Depends(get_db),
    ) -> User:
        payload = decode_token(credentials.credentials)
        if payload.get("type") != "access":
            raise HTTPException(status_code=status.HTTP_401_UNAUTHORIZED,
                                detail="Invalid token type")

        user_id = int(payload["sub"])
        result = await db.execute(select(User).where(User.id == user_id,
            User.is_active.is_(True)))
        user = result.scalar_one_or_none()
        if user is None:
            raise HTTPException(status_code=status.HTTP_401_UNAUTHORIZED, detail="User
            not found")
        return user

def require_role(*roles: UserRole):
    def checker(current_user: User = Depends(get_current_user)) -> User:
        if current_user.role not in roles:
            raise HTTPException(status_code=status.HTTP_403_FORBIDDEN,
                                detail="Insufficient permissions")
        return current_user
    return checker

```

```

=====
ФАЙЛ: ml_service/service/server.py
Описание: ML-сервис (FastAPI) – классификация, предсказание срывов, обучение
=====

```

```

from __future__ import annotations

import os
import time

import numpy as np
import structlog
from fastapi import FastAPI, HTTPException
from fastapi.responses import Response
from pydantic import BaseModel

from service.metrics import CONTENT_TYPE_LATEST, ML_INFERENCE_DURATION,
ML_TRAINING_DURATION, generate_latest

from service.data.dataset import get_sample_dataset
from service.data.fair_mast_loader import load_disruption_labels, load_shots,
make_disruption_labels
from service.data.metrics import compute_metrics, format_confusion_matrix
from service.data.preprocessing import prepare_batch, prepare_features
from service.training.trainer import create_model, load_model

```

```

logger = structlog.get_logger()

app = FastAPI(title="Tokamak ML Service", version="1.0.0")

# Cache for loaded models
_model_cache: dict[str, object] = {}

class PredictRequest(BaseModel):
    data: dict
    model_path: str
    model_type: str
    threshold: float = 0.7

class TrainRequest(BaseModel):
    model_type: str
    task: str
    hyperparameters: dict = {}
    shot_ids: list[int] = []
    run_id: int | None = None

@app.get("/health")
async def health():
    return {"status": "healthy", "service": "ml"}

@app.get("/metrics")
async def metrics():
    return Response(content=generate_latest(), media_type=CONTENT_TYPE_LATEST)

@app.post("/api/v1/predict/classify")
async def classify(request: PredictRequest):
    try:
        # Prepare features
        features = prepare_features(request.data.get("signals", {}))

        # Load or get cached model
        model = _get_model(request.model_type, request.model_path)

        start = time.perf_counter()
        prediction = model.predict(features)
        proba = model.predict_proba(features)
        inference_ms = (time.perf_counter() - start) * 1000

        label = "stable" if prediction[0] == 0 else "unstable"
        confidence = float(proba[0].max())

        logger.info("classification_done", label=label, confidence=confidence,
inference_ms=inference_ms)

    return {
        "label": label,
        "confidence": confidence,
        "inference_time_ms": inference_ms,
    }

```

```

    }
except Exception as e:
    logger.error("classification_failed", error=str(e))
    raise HTTPException(status_code=500, detail=str(e))

@app.post("/api/v1/predict/disruption")
async def predict_disruption(request: PredictRequest):
    try:
        features = prepare_features(request.data.get("signals", {}))
        model = _get_model(request.model_type, request.model_path)

        start = time.perf_counter()
        proba = model.predict_proba(features)
        inference_ms = (time.perf_counter() - start) * 1000

        probabilities = proba[:, 1].tolist() if proba.ndim > 1 else proba.tolist()

        # Generate timestamps (normalized)
        n_steps = len(probabilities)
        timestamps = list(np.linspace(0, 1, n_steps))

        # Check if warning should be issued
        max_prob = max(probabilities)
        warning_issued = max_prob > request.threshold
        warning_time_ms = None
        if warning_issued:
            warning_idx = next(i for i, p in enumerate(probabilities) if p >
request.threshold)
            warning_time_ms = timestamps[warning_idx] * 1000

        return {
            "timestamps": timestamps,
            "probabilities": probabilities,
            "warning_issued": warning_issued,
            "warning_time_ms": warning_time_ms,
            "max_probability": max_prob,
            "inference_time_ms": inference_ms,
        }
    except Exception as e:
        logger.error("disruption_prediction_failed", error=str(e))
        raise HTTPException(status_code=500, detail=str(e))

@app.post("/api/v1/train")
async def train(request: TrainRequest):
    try:
        logger.info("training_started", model_type=request.model_type,
task=request.task)

        model = create_model(request.model_type, request.hyperparameters or None)

        seq_len = request.hyperparameters.get("sequence_length", 200)
        n_features = request.hyperparameters.get("input_size", 39)

        if request.shot_ids:
            # Load real data from FAIR-MAST
            shots_data, loaded_ids = load_shots(request.shot_ids)

```

```

        if not shots_data:
            raise HTTPException(
                status_code=404,
                detail="No valid shots could be loaded from the provided
shot_ids",
            )
        X = prepare_batch(
            [{"signals": s} for s in shots_data],
            sequence_length=seq_len,
        )
        # Load real disruption labels from FAIR-MAST
        disruption_times = load_disruption_labels(loaded_ids)
        y = make_disruption_labels(loaded_ids, disruption_times)

        logger.info(
            "disruption_labels_loaded",
            n_disrupted=int(y.sum()),
            n_stable=int(len(y) - y.sum()),
        )

        from service.data.dataset import build_dataset
        dataset = build_dataset(X, y)
    else:
        # Use synthetic sample data for testing / development
        dataset = get_sample_dataset(
            n_samples=200,
            seq_len=seq_len,
            n_features=n_features,
            task=request.task,
        )

    # Train the model
    if request.model_type == "random_forest":
        X_tr, y_tr = dataset.flat_train()
        X_v, y_v = dataset.flat_val()
        X_te, y_te = dataset.flat_test()
    else:
        X_tr, y_tr = dataset.X_train, dataset.y_train
        X_v, y_v = dataset.X_val, dataset.y_val
        X_te, y_te = dataset.X_test, dataset.y_test

    train_metrics = model.fit(X_tr, y_tr, X_v, y_v)

    # Compute proper test metrics
    y_pred = model.predict(X_te)
    y_proba = model.predict_proba(X_te)
    test_metrics = compute_metrics(y_te, y_pred, y_proba)
    cm = format_confusion_matrix(y_te, y_pred)

    # Save model to persistent directory
    save_dir = "/tmp/tokamak_models"
    os.makedirs(save_dir, exist_ok=True)
    ext = ".joblib" if request.model_type == "random_forest" else ".pt"
    run_tag = f"_run{request.run_id}" if request.run_id else ""
    save_path = os.path.join(
        save_dir, f"{request.model_type}_{request.task}{run_tag}{ext}"
    )
    model.save(save_path)

```

```

        logger.info(
            "training_completed",
            model_type=request.model_type,
            test_metrics=test_metrics,
        )

    return {
        "status": "completed",
        "train_metrics": train_metrics,
        "test_metrics": test_metrics,
        "confusion_matrix": cm,
        "model_path": save_path,
        "metadata": model.metadata(),
    }
except HTTPException:
    raise
except Exception as e:
    logger.error("training_failed", error=str(e))
    raise HTTPException(status_code=500, detail=str(e))

```

```

def _get_model(model_type: str, model_path: str):
    """Load model with caching."""
    cache_key = f"{model_type}:{model_path}"
    if cache_key not in _model_cache:
        _model_cache[cache_key] = load_model(model_type, model_path)
    return _model_cache[cache_key]

```

=====

ФАЙЛ: ml_service/service/models/interface.py

Описание: Абстрактный интерфейс ModelInterface (протокол для всех ML-моделей)

=====

```

from __future__ import annotations

```

```

from abc import ABC, abstractmethod
from typing import Any

```

```

import numpy as np

```

```

class ModelInterface(ABC):

```

```

    """Unified interface for all ML models (ADR-004).

```

```

    All three models (Random Forest, bi-LSTM+attention, Transformer)
    implement this interface for interchangeability in the training
    and inference pipeline.
    """

```

```

    @abstractmethod

```

```

    def fit(
        self,
        X_train: np.ndarray,
        y_train: np.ndarray,
        X_val: np.ndarray | None = None,
        y_val: np.ndarray | None = None,

```



```

        hyperparams: dict | None = None,
        progress_callback: Any = None,
    ) -> dict:
        """Train the model.

        Returns dict with training metrics (loss, accuracy, auc, etc.).
        """
        ...

    @abstractmethod
    def predict(self, X: np.ndarray) -> np.ndarray:
        """Return class labels (classification) or disruption probabilities over
        time."""
        ...

    @abstractmethod
    def predict_proba(self, X: np.ndarray) -> np.ndarray:
        """Return probability estimates for AUC-ROC and calibration curves."""
        ...

    @abstractmethod
    def save(self, path: str) -> None:
        """Serialize model to file (joblib for RF, TorchScript for PyTorch
        models)."""
        ...

    @abstractmethod
    def load(self, path: str) -> "ModelInterface":
        """Load model from file."""
        ...

    @abstractmethod
    def metadata(self) -> dict:
        """Return model metadata: architecture, hyperparameters, training date,
        validation metrics, dataset hash."""
        ...

```

```

=====
ФАЙЛ: ml_service/service/models/random_forest.py
Описание: Модель Random Forest (базовая модель, scikit-learn)
=====

```

```

from __future__ import annotations

import hashlib
import json
from datetime import datetime, timezone
from typing import Any

import joblib
import numpy as np
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, f1_score, roc_auc_score

from service.models.interface import ModelInterface

```

```

class RandomForestModel(ModelInterface):
    """Random Forest baseline model (ADR-004 §3.2.1).

    Used as baseline for classification (FR-004).
    CPU inference <= 5ms. No GPU required.
    """

    RF_PARAMS = {
        "n_estimators", "max_depth", "min_samples_split", "min_samples_leaf",
        "class_weight", "max_features", "random_state", "n_jobs",
        "criterion", "min_weight_fraction_leaf", "max_leaf_nodes",
        "min_impurity_decrease", "bootstrap", "oob_score", "verbose",
        "warm_start", "ccp_alpha", "max_samples",
    }

    def __init__(self, hyperparams: dict | None = None):
        defaults = {
            "n_estimators": 200,
            "max_depth": 20,
            "min_samples_split": 5,
            "min_samples_leaf": 2,
            "class_weight": "balanced",
            "max_features": "sqrt",
            "random_state": 42,
            "n_jobs": -1,
        }
        if hyperparams:
            # Only keep params that RF actually accepts
            filtered = {k: v for k, v in hyperparams.items() if k in
self.RF_PARAMS}
            defaults.update(filtered)
        self.hyperparams = defaults
        self.model = RandomForestClassifier(**self.hyperparams)
        self.metrics: dict = {}
        self.trained_at: str | None = None
        self.dataset_hash: str | None = None

    def fit(
        self,
        X_train: np.ndarray,
        y_train: np.ndarray,
        X_val: np.ndarray | None = None,
        y_val: np.ndarray | None = None,
        hyperparams: dict | None = None,
        progress_callback: Any = None,
    ) -> dict:
        if hyperparams:
            self.hyperparams.update(hyperparams)
            self.model = RandomForestClassifier(**self.hyperparams)

        self.dataset_hash = hashlib.sha256(
            X_train.tobytes() + y_train.tobytes()
        ).hexdigest()[:16]

        self.model.fit(X_train, y_train)
        self.trained_at = datetime.now(timezone.utc).isoformat()

        self.metrics = {"train_accuracy": accuracy_score(y_train,

```

```

self.model.predict(X_train))}

    if X_val is not None and y_val is not None:
        y_pred = self.model.predict(X_val)
        y_proba = self.model.predict_proba(X_val)

        self.metrics.update({
            "val_accuracy": accuracy_score(y_val, y_pred),
            "val_f1": f1_score(y_val, y_pred, average="weighted"),
        })
        if y_proba.shape[1] == 2:
            self.metrics["val_auc_roc"] = roc_auc_score(y_val, y_proba[:, 1])

    return self.metrics

def predict(self, X: np.ndarray) -> np.ndarray:
    return self.model.predict(X)

def predict_proba(self, X: np.ndarray) -> np.ndarray:
    return self.model.predict_proba(X)

def save(self, path: str) -> None:
    artifact = {
        "model": self.model,
        "hyperparams": self.hyperparams,
        "metrics": self.metrics,
        "trained_at": self.trained_at,
        "dataset_hash": self.dataset_hash,
    }
    joblib.dump(artifact, path)

def load(self, path: str) -> "RandomForestModel":
    artifact = joblib.load(path)
    self.model = artifact["model"]
    self.hyperparams = artifact["hyperparams"]
    self.metrics = artifact["metrics"]
    self.trained_at = artifact["trained_at"]
    self.dataset_hash = artifact["dataset_hash"]
    return self

def metadata(self) -> dict:
    return {
        "architecture": "RandomForest",
        "hyperparameters": self.hyperparams,
        "trained_at": self.trained_at,
        "metrics": self.metrics,
        "dataset_hash": self.dataset_hash,
        "feature_importances": (
            self.model.feature_importances_.tolist()
            if hasattr(self.model, "feature_importances_")
            else None
        ),
    }
}

```

=====

ФАЙЛ: ml_service/service/models/lstm_attention.py

Описание: Модель bi-LSTM с multi-head attention (PyTorch)

```

=====

from __future__ import annotations

import hashlib
from datetime import datetime, timezone
from typing import Any

import numpy as np
import torch
import torch.nn as nn

from service.models.interface import ModelInterface

class AttentionLayer(nn.Module):
    """Multi-head attention over LSTM outputs."""

    def __init__(self, hidden_size: int, num_heads: int = 4):
        super().__init__()
        self.attention = nn.MultiheadAttention(
            embed_dim=hidden_size, num_heads=num_heads, batch_first=True
        )
        self.norm = nn.LayerNorm(hidden_size)

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        attn_out, _ = self.attention(x, x, x)
        return self.norm(x + attn_out)

class BiLSTMAttentionNet(nn.Module):
    """Bidirectional LSTM with multi-head attention (ADR-004 §3.2.2).

    Based on HDL (Zhu et al., NC 2020) and FRNN (Kates-Harbeck et al., Nature
    2019).
    """

    def __init__(
        self,
        input_size: int = 39,
        hidden_size: int = 128,
        num_layers: int = 2,
        num_heads: int = 4,
        dropout: float = 0.3,
        bidirectional: bool = True,
    ):
        super().__init__()
        self.lstm = nn.LSTM(
            input_size=input_size,
            hidden_size=hidden_size,
            num_layers=num_layers,
            batch_first=True,
            bidirectional=bidirectional,
            dropout=dropout if num_layers > 1 else 0.0,
        )
        lstm_output_size = hidden_size * (2 if bidirectional else 1)
        self.attention = AttentionLayer(lstm_output_size, num_heads)
        self.classifier = nn.Sequential(

```

```

        nn.Linear(lstm_output_size, 64),
        nn.ReLU(),
        nn.Dropout(dropout),
        nn.Linear(64, 1),
    )

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        lstm_out, _ = self.lstm(x)
        attn_out = self.attention(lstm_out)
        # Global average pooling over time
        pooled = attn_out.mean(dim=1)
        return self.classifier(pooled).squeeze(-1)

```

```

class LSTMAttentionModel(ModelInterface):

```

```

    """bi-LSTM+attention model for classification and disruption prediction.

```

```

    Primary production model (ADR-004). GPU inference <= 30ms on T4.
    """

```

```

    def __init__(self, hyperparams: dict | None = None):
        self.hyperparams = hyperparams or {
            "input_size": 39,
            "hidden_size": 128,
            "num_layers": 2,
            "bidirectional": True,
            "dropout": 0.3,
            "attention_heads": 4,
            "sequence_length": 200,
            "learning_rate": 1e-3,
            "batch_size": 32,
            "epochs": 50,
            "early_stopping_patience": 10,
            "weight_decay": 1e-4,
        }
        self.device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
        self.model = BiLSTMAttentionNet(
            input_size=self.hyperparams["input_size"],
            hidden_size=self.hyperparams["hidden_size"],
            num_layers=self.hyperparams["num_layers"],
            num_heads=self.hyperparams["attention_heads"],
            dropout=self.hyperparams["dropout"],
            bidirectional=self.hyperparams["bidirectional"],
        ).to(self.device)
        self.metrics: dict = {}
        self.trained_at: str | None = None
        self.dataset_hash: str | None = None

```

```

    def fit(
        self,
        X_train: np.ndarray,
        y_train: np.ndarray,
        X_val: np.ndarray | None = None,
        y_val: np.ndarray | None = None,
        hyperparams: dict | None = None,
        progress_callback: Any = None,
    ) -> dict:
        if hyperparams:

```

```

        self.hyperparams.update(hyperparams)

self.dataset_hash = hashlib.sha256(X_train.tobytes()[:10000]).hexdigest()
[:16]

X_t = torch.FloatTensor(X_train).to(self.device)
y_t = torch.FloatTensor(y_train).to(self.device)

dataset = torch.utils.data.TensorDataset(X_t, y_t)
loader = torch.utils.data.DataLoader(
    dataset, batch_size=self.hyperparams["batch_size"], shuffle=True
)

optimizer = torch.optim.AdamW(
    self.model.parameters(),
    lr=self.hyperparams["learning_rate"],
    weight_decay=self.hyperparams["weight_decay"],
)
scheduler = torch.optim.lr_scheduler.ReduceLROnPlateau(optimizer,
factor=0.5, patience=5)

# Class weights for imbalanced data
pos_weight = torch.tensor([(y_train == 0).sum() / max((y_train == 1).sum(),
1)]).to(self.device)
criterion = nn.BCEWithLogitsLoss(pos_weight=pos_weight)

best_val_loss = float("inf")
patience_counter = 0
history = {"train_loss": [], "val_loss": []}

self.model.train()
for epoch in range(self.hyperparams["epochs"]):
    epoch_loss = 0.0
    for X_batch, y_batch in loader:
        optimizer.zero_grad()
        output = self.model(X_batch)
        loss = criterion(output, y_batch)
        loss.backward()
        torch.nn.utils.clip_grad_norm_(self.model.parameters(),
max_norm=1.0)
        optimizer.step()
        epoch_loss += loss.item()

    avg_loss = epoch_loss / len(loader)
    history["train_loss"].append(avg_loss)

    # Validation
    val_loss = avg_loss
    if X_val is not None and y_val is not None:
        self.model.eval()
        with torch.no_grad():
            X_v = torch.FloatTensor(X_val).to(self.device)
            y_v = torch.FloatTensor(y_val).to(self.device)
            val_output = self.model(X_v)
            val_loss = criterion(val_output, y_v).item()
        history["val_loss"].append(val_loss)
    self.model.train()

```

```

        scheduler.step(val_loss)

        if val_loss < best_val_loss:
            best_val_loss = val_loss
            patience_counter = 0
        else:
            patience_counter += 1
            if patience_counter >= self.hyperparams["early_stopping_patience"]:
                break

        if progress_callback:
            progress_callback(epoch + 1, self.hyperparams["epochs"], avg_loss,
val_loss)

        self.trained_at = datetime.now(timezone.utc).isoformat()
        self.metrics = {
            "final_train_loss": history["train_loss"][-1],
            "best_val_loss": best_val_loss,
            "epochs_trained": len(history["train_loss"]),
            "history": history,
        }
        return self.metrics

def predict(self, X: np.ndarray) -> np.ndarray:
    self.model.eval()
    with torch.no_grad():
        X_t = torch.FloatTensor(X).to(self.device)
        output = torch.sigmoid(self.model(X_t))
        return (output.cpu().numpy() > 0.5).astype(int)

def predict_proba(self, X: np.ndarray) -> np.ndarray:
    self.model.eval()
    with torch.no_grad():
        X_t = torch.FloatTensor(X).to(self.device)
        proba = torch.sigmoid(self.model(X_t)).cpu().numpy()
        return np.column_stack([1 - proba, proba])

def save(self, path: str) -> None:
    torch.save({
        "model_state_dict": self.model.state_dict(),
        "hyperparams": self.hyperparams,
        "metrics": self.metrics,
        "trained_at": self.trained_at,
        "dataset_hash": self.dataset_hash,
    }, path)

def load(self, path: str) -> "LSTMAttentionModel":
    checkpoint = torch.load(path, map_location=self.device, weights_only=False)
    self.hyperparams = checkpoint["hyperparams"]
    self.model = BiLSTMAttentionNet(
        input_size=self.hyperparams["input_size"],
        hidden_size=self.hyperparams["hidden_size"],
        num_layers=self.hyperparams["num_layers"],
        num_heads=self.hyperparams["attention_heads"],
        dropout=self.hyperparams["dropout"],
        bidirectional=self.hyperparams["bidirectional"],
    ).to(self.device)
    self.model.load_state_dict(checkpoint["model_state_dict"])

```

```

        self.metrics = checkpoint["metrics"]
        self.trained_at = checkpoint["trained_at"]
        self.dataset_hash = checkpoint["dataset_hash"]
        return self

    def metadata(self) -> dict:
        return {
            "architecture": "BiLSTM_Attention",
            "hyperparameters": self.hyperparams,
            "trained_at": self.trained_at,
            "metrics": self.metrics,
            "dataset_hash": self.dataset_hash,
            "device": str(self.device),
            "total_parameters": sum(p.numel() for p in self.model.parameters()),
        }

```

```

=====
ФАЙЛ: ml_service/service/models/transformer.py
Описание: Autoregressive Transformer с RoPE (PyTorch)
=====

```

```

from __future__ import annotations

```

```

import hashlib
import math
from datetime import datetime, timezone
from typing import Any

```

```

import numpy as np
import torch
import torch.nn as nn

```

```

from service.models.interface import ModelInterface

```

```

class RotaryPositionalEncoding(nn.Module):
    """Rotary Position Embedding (RoPE)."""

```

```

    def __init__(self, d_model: int, max_len: int = 5000):
        super().__init__()
        position = torch.arange(max_len).unsqueeze(1).float()
        div_term = torch.exp(torch.arange(0, d_model, 2).float() * (-
math.log(10000.0) / d_model))
        pe = torch.zeros(max_len, d_model)
        pe[:, 0::2] = torch.sin(position * div_term)
        pe[:, 1::2] = torch.cos(position * div_term)
        self.register_buffer("pe", pe.unsqueeze(0))

```

```

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        return x + self.pe[:, : x.size(1)]

```

```

class AutoregressiveTransformer(nn.Module):
    """Autoregressive Transformer for disruption prediction (ADR-004 §3.2.3).

```

```

    Based on Spangher et al. (arXiv:2401.00051).
    Causal attention mask for autoregressive prediction.

```



```

"""

def __init__(
    self,
    input_size: int = 39,
    d_model: int = 128,
    num_heads: int = 8,
    num_layers: int = 4,
    dim_feedforward: int = 512,
    dropout: float = 0.1,
    max_seq_len: int = 1000,
):
    super().__init__()
    self.input_proj = nn.Linear(input_size, d_model)
    self.pos_encoding = RotaryPositionalEncoding(d_model, max_seq_len)
    self.dropout = nn.Dropout(dropout)

    encoder_layer = nn.TransformerEncoderLayer(
        d_model=d_model,
        nhead=num_heads,
        dim_feedforward=dim_feedforward,
        dropout=dropout,
        batch_first=True,
        activation="gelu",
    )
    self.transformer = nn.TransformerEncoder(encoder_layer,
num_layers=num_layers)
    self.output_head = nn.Sequential(
        nn.Linear(d_model, d_model // 2),
        nn.GELU(),
        nn.Dropout(dropout),
        nn.Linear(d_model // 2, 1),
    )

    def _generate_causal_mask(self, sz: int, device: torch.device) -> torch.Tensor:
        return torch.triu(torch.ones(sz, sz, device=device) * float("-inf"),
diagonal=1)

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        x = self.input_proj(x)
        x = self.pos_encoding(x)
        x = self.dropout(x)

        mask = self._generate_causal_mask(x.size(1), x.device)
        x = self.transformer(x, mask=mask)

        # Output per-timestep predictions
        return self.output_head(x).squeeze(-1)

class TransformerModel(ModelInterface):
    """Autoregressive Transformer for disruption prediction.

    Experimental model (ADR-004). GPU inference <= 50ms on T4.
    """

    def __init__(self, hyperparams: dict | None = None):
        self.hyperparams = hyperparams or {

```

```

        "input_size": 39,
        "d_model": 128,
        "num_heads": 8,
        "num_layers": 4,
        "dim_feedforward": 512,
        "dropout": 0.1,
        "sequence_length": 500,
        "learning_rate": 5e-4,
        "warmup_steps": 1000,
        "batch_size": 16,
        "epochs": 40,
        "early_stopping_patience": 8,
        "weight_decay": 1e-4,
    }
    self.device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
    self.model = AutoregressiveTransformer(
        input_size=self.hyperparams["input_size"],
        d_model=self.hyperparams["d_model"],
        num_heads=self.hyperparams["num_heads"],
        num_layers=self.hyperparams["num_layers"],
        dim_feedforward=self.hyperparams["dim_feedforward"],
        dropout=self.hyperparams["dropout"],
        max_seq_len=self.hyperparams["sequence_length"],
    ).to(self.device)
    self.metrics: dict = {}
    self.trained_at: str | None = None
    self.dataset_hash: str | None = None

```

```

def fit(
    self,
    X_train: np.ndarray,
    y_train: np.ndarray,
    X_val: np.ndarray | None = None,
    y_val: np.ndarray | None = None,
    hyperparams: dict | None = None,
    progress_callback: Any = None,
) -> dict:
    if hyperparams:
        self.hyperparams.update(hyperparams)

```

```

    self.dataset_hash = hashlib.sha256(X_train.tobytes()[:10000]).hexdigest()

```

[:16]

```

X_t = torch.FloatTensor(X_train).to(self.device)
y_t = torch.FloatTensor(y_train).to(self.device)

dataset = torch.utils.data.TensorDataset(X_t, y_t)
loader = torch.utils.data.DataLoader(
    dataset, batch_size=self.hyperparams["batch_size"], shuffle=True
)

optimizer = torch.optim.AdamW(
    self.model.parameters(),
    lr=self.hyperparams["learning_rate"],
    weight_decay=self.hyperparams["weight_decay"],
)

# Cosine annealing with warmup

```

```

def lr_lambda(step):
    if step < self.hyperparams["warmup_steps"]:
        return step / max(self.hyperparams["warmup_steps"], 1)
    progress = (step - self.hyperparams["warmup_steps"]) / max(
        len(loader) * self.hyperparams["epochs"] -
self.hyperparams["warmup_steps"], 1
    )
    return 0.5 * (1 + math.cos(math.pi * progress))

scheduler = torch.optim.lr_scheduler.LambdaLR(optimizer, lr_lambda)
criterion = nn.BCEWithLogitsLoss()

best_val_loss = float("inf")
patience_counter = 0
history = {"train_loss": [], "val_loss": []}

self.model.train()
for epoch in range(self.hyperparams["epochs"]):
    epoch_loss = 0.0
    for X_batch, y_batch in loader:
        optimizer.zero_grad()
        output = self.model(X_batch)
        # Use last timestep prediction for classification
        if output.dim() > 1:
            output = output[:, -1]
        loss = criterion(output, y_batch)
        loss.backward()
        torch.nn.utils.clip_grad_norm_(self.model.parameters(),
max_norm=1.0)
        optimizer.step()
        scheduler.step()
        epoch_loss += loss.item()

    avg_loss = epoch_loss / len(loader)
    history["train_loss"].append(avg_loss)

    val_loss = avg_loss
    if X_val is not None and y_val is not None:
        self.model.eval()
        with torch.no_grad():
            X_v = torch.FloatTensor(X_val).to(self.device)
            y_v = torch.FloatTensor(y_val).to(self.device)
            val_output = self.model(X_v)
            if val_output.dim() > 1:
                val_output = val_output[:, -1]
            val_loss = criterion(val_output, y_v).item()
        history["val_loss"].append(val_loss)
        self.model.train()

    if val_loss < best_val_loss:
        best_val_loss = val_loss
        patience_counter = 0
    else:
        patience_counter += 1
        if patience_counter >= self.hyperparams["early_stopping_patience"]:
            break

    if progress_callback:

```

```

        progress_callback(epoch + 1, self.hyperparams["epochs"], avg_loss,
val_loss)

        self.trained_at = datetime.now(timezone.utc).isoformat()
        self.metrics = {
            "final_train_loss": history["train_loss"][-1],
            "best_val_loss": best_val_loss,
            "epochs_trained": len(history["train_loss"]),
        }
        return self.metrics

def predict(self, X: np.ndarray) -> np.ndarray:
    self.model.eval()
    with torch.no_grad():
        X_t = torch.FloatTensor(X).to(self.device)
        output = torch.sigmoid(self.model(X_t))
        if output.dim() > 1:
            output = output[:, -1]
        return (output.cpu().numpy() > 0.5).astype(int)

def predict_proba(self, X: np.ndarray) -> np.ndarray:
    self.model.eval()
    with torch.no_grad():
        X_t = torch.FloatTensor(X).to(self.device)
        output = torch.sigmoid(self.model(X_t))
        if output.dim() > 1:
            output = output[:, -1]
        proba = output.cpu().numpy()
        return np.column_stack([1 - proba, proba])

def save(self, path: str) -> None:
    torch.save({
        "model_state_dict": self.model.state_dict(),
        "hyperparams": self.hyperparams,
        "metrics": self.metrics,
        "trained_at": self.trained_at,
        "dataset_hash": self.dataset_hash,
    }, path)

def load(self, path: str) -> "TransformerModel":
    checkpoint = torch.load(path, map_location=self.device, weights_only=False)
    self.hyperparams = checkpoint["hyperparams"]
    self.model = AutoregressiveTransformer(
        input_size=self.hyperparams["input_size"],
        d_model=self.hyperparams["d_model"],
        num_heads=self.hyperparams["num_heads"],
        num_layers=self.hyperparams["num_layers"],
        dim_feedforward=self.hyperparams["dim_feedforward"],
        dropout=self.hyperparams["dropout"],
        max_seq_len=self.hyperparams["sequence_length"],
    ).to(self.device)
    self.model.load_state_dict(checkpoint["model_state_dict"])
    self.metrics = checkpoint["metrics"]
    self.trained_at = checkpoint["trained_at"]
    self.dataset_hash = checkpoint["dataset_hash"]
    return self

def metadata(self) -> dict:

```

```

    return {
      "architecture": "AutoregressiveTransformer",
      "hyperparameters": self.hyperparams,
      "trained_at": self.trained_at,
      "metrics": self.metrics,
      "dataset_hash": self.dataset_hash,
      "device": str(self.device),
      "total_parameters": sum(p.numel() for p in self.model.parameters()),
    }

```

=====

ФАЙЛ: frontend/src/App.tsx

Описание: Корневой компонент React-приложения, маршрутизация

=====

```

import { BrowserRouter, Navigate, Route, Routes } from 'react-router-dom';
import { useAuthStore } from './store/authStore';
import Layout from './components/Layout';
import ToastContainer from './components/Toast';
import LoginPage from './pages/LoginPage';
import DashboardPage from './pages/DashboardPage';
import ExperimentPage from './pages/ExperimentPage';
import TrainingPage from './pages/TrainingPage';
import BatchPage from './pages/BatchPage';
import SettingsPage from './pages/SettingsPage';

```

```

function ProtectedRoute({ children }: { children: React.ReactNode }) {
  const isAuthenticated = useAuthStore((s) => s.isAuthenticated);
  return isAuthenticated ? <>{children}</> : <Navigate to="/login" />;
}

```

```

export default function App() {
  return (
    <BrowserRouter>
      <ToastContainer />
      <Routes>
        <Route path="/login" element={<LoginPage />} />
        <Route
          path="/"
          element={
            <ProtectedRoute>
              <Layout />
            </ProtectedRoute>
          }
        />
        <Route index element={<DashboardPage />} />
        <Route path="experiment/:id" element={<ExperimentPage />} />
        <Route path="training" element={<TrainingPage />} />
        <Route path="batch" element={<BatchPage />} />
        <Route path="settings" element={<SettingsPage />} />
      </Routes>
    </BrowserRouter>
  );
}

```

=====

ФАЙЛ: frontend/src/pages/DashboardPage.tsx

Описание: Главная страница – статистика, загрузка данных, сравнение моделей

=====

```
import { useEffect, useState } from 'react';
import { useNavigate } from 'react-router-dom';
import Plot from 'react-plotly.js';
import api from '../api/client';
import ShotLoader from '../components/ShotLoader';
import type { Experiment, MLModel } from '../types';

export default function DashboardPage() {
  const [experiments, setExperiments] = useState<Experiment[]>([]);
  const [models, setModels] = useState<MLModel[]>([]);
  const [loading, setLoading] = useState(true);
  const navigate = useNavigate();

  useEffect(() => {
    api
      .get('/experiments/')
      .then((res) => setExperiments(res.data.experiments))
      .catch(() => {})
      .finally(() => setLoading(false));

    api
      .get('/models/')
      .then((res) => setModels(res.data))
      .catch(() => {});
  }, []);

  const handleLoaded = (exp: Experiment) => {
    setExperiments((prev) => [exp, ...prev]);
    navigate(`/experiment/${exp.id}`);
  };

  const preprocessedCount = experiments.filter((e) => e.status ===
'preprocessed').length;
  const totalSignals = experiments.reduce(
    (sum, e) => sum + ((e.metadata_json?.signal_count as number) || 0),
    0
  );

  const statusBadge = (status: string) => {
    const cls =
      status === 'preprocessed'
        ? 'badge-success'
        : status === 'error'
        ? 'badge-danger'
        : 'badge-warning';
    return <span className={`badge ${cls}`}>{status}</span>;
  };

  return (
    <div>
      <h2 className="mb-lg">Dashboard</h2>

      {/* Summary Stats */}
```

```

<div className="grid-3 mb-lg">
  <div className="stat-card">
    <div className="stat-icon stat-icon-primary">{'\u26A1'}</div>
    <div>
      <div className="stat-value">{experiments.length}</div>
      <div className="stat-label">Экспериментов</div>
    </div>
  </div>
  <div className="stat-card">
    <div className="stat-icon stat-icon-success">{'\u2713'}</div>
    <div>
      <div className="stat-value">{preprocessedCount}</div>
      <div className="stat-label">Обработано</div>
    </div>
  </div>
  <div className="stat-card">
    <div className="stat-icon stat-icon-warning">{'\u223F'}</div>
    <div>
      <div className="stat-value">{totalSignals}</div>
      <div className="stat-label">Сигналов</div>
    </div>
  </div>
</div>

```

```

<ShotLoader onLoaded={handleLoaded} />

```

```

{/* Model Comparison */}

```

```

{models.length > 0 && (
  <div className="card mb-lg">
    <div className="card-header">
      <h3>Сравнение моделей</h3>
    </div>

```

```

    <table className="table" style={{ marginBottom: 20 }}>

```

```

      <thead>

```

```

        <tr>

```

```

          <th>Модель</th>

```

```

          <th>Тип</th>

```

```

          <th>Accuracy</th>

```

```

          <th>F1</th>

```

```

          <th>AUC</th>

```

```

          <th>Статус</th>

```

```

        </tr>

```

```

      </thead>

```

```

      <tbody>

```

```

        {models.map((m) => (

```

```

          <tr key={m.id}>

```

```

            <td className="font-semibold">{m.name}</td>

```

```

            <td>

```

```

              <span className="badge badge-info">{m.model_type}</span>

```

```

            </td>

```

```

            <td>{m.metrics_json?.accuracy != null ?

```

```

m.metrics_json.accuracy.toFixed(4) : '\u2014'}</td>

```

```

            <td>{m.metrics_json?.f1 != null ? m.metrics_json.f1.toFixed(4) :

```

```

'\u2014'}</td>

```

```

            <td>{m.metrics_json?.auc != null ?

```

```

m.metrics_json.auc.toFixed(4) : '\u2014'}</td>

```

```

            <td>

```

```

        <span className={`badge ${m.is_active ? 'badge-success' :
'badge-neutral'}}`>
            {m.is_active ? 'Активна' : 'Неактивна'}
        </span>
    </td>
</tr>
</tbody>
</table>

{/* AUC bar chart */}
{models.some((m) => m.metrics_json?.auc != null) && (
    <Plot
        data={[
            {
                x: models.filter((m) => m.metrics_json?.auc != null).map((m) =>
m.name),
                y: models.filter((m) => m.metrics_json?.auc != null).map((m) =>
m.metrics_json!.auc),
                type: 'bar',
                marker: {
                    color: models
                        .filter((m) => m.metrics_json?.auc != null)
                        .map((m) => (m.is_active ? '#2ecc71' : '#3498db')),
                },
                hovertemplate: '%{x}<br>AUC: %{y:.4f}<extra></extra>',
            },
        ]}
        layout={{
            title: 'Сравнение AUC моделей',
            yaxis: { title: 'AUC', range: [0, 1] },
            height: 350,
            margin: { l: 60, r: 30, t: 50, b: 80 },
        }}
        config={{ responsive: true, displayModeBar: false }}
        style={{ width: '100%' }}
    />
)}
</div>
)}

<div className="card">
    <div className="card-header">
        <h3>Загруженные эксперименты</h3>
    </div>

    {loading ? (
        <div className="loading-overlay">
            <span className="spinner spinner-lg" />
            Загрузка данных...
        </div>
    ) : experiments.length === 0 ? (
        <div className="empty-state">
            <div className="empty-state-icon">{'\u269B'}</div>
            <div className="empty-state-text">Нет загруженных экспериментов</div>
            <div className="empty-state-hint">
                Загрузите данные выстрела, указав Shot ID выше
            </div>
        </div>
    ) : experiments.length > 0 ? (
        <div className="experiments">
            {experiments.map((exp) => (
                <div className="experiment-card">
                    <div>{exp.name}</div>
                    <div>{exp.metrics_json?.auc}</div>
                </div>
            ))}
        </div>
    ) : null}
    </div>

```



```

    </div>
  ) : (
    <table className="table">
      <thead>
        <tr>
          <th>Shot ID</th>
          <th>Источник</th>
          <th>Статус</th>
          <th>Дата</th>
          <th>Сигналов</th>
          <th></th>
        </tr>
      </thead>
      <tbody>
        {experiments.map((exp) => (
          <tr key={exp.id}>
            <td className="font-semibold">{exp.shot_id}</td>
            <td>
              <span className="badge badge-info">{exp.source.toUpperCase()}</span>
            </td>
            <td>{statusBadge(exp.status)}</td>
            <td className="text-sm text-secondary">
              {new Date(exp.loaded_at).toLocaleString('ru')}
            </td>
            <td>{(exp.metadata_json?.signal_count as number) ?? '\u2014'}</td>
            <td>
              <button
                className="btn btn-primary btn-sm"
                onClick={() => navigate(`/experiment/${exp.id}`)}
              >
                Открыть
              </button>
            </td>
          </tr>
        ))}
      </tbody>
    </table>
  )}
</div>
</div>
);
}

```

```

=====
КОНЕЦ ДЕПОНИРУЕМЫХ МАТЕРИАЛОВ
Всего файлов: 10
=====

```

Аудиовизуальные отображения программы для ЭВМ «Tokamak Analysis»

ИИ-система для анализа экспериментальных данных токамаков

Правообладатель: Киселев Фёдор Сергеевич

Год создания: 2026

Скриншоты веб-интерфейса: 9 экранов

Страница входа в систему



Tokamak Analysis

НИЯУ МИФИ — ИИ-система анализа данных
токамаков

Email

admin@mephi.ru

Пароль

.....



Войти

[Нет аккаунта? Зарегистрироваться](#)

НАВИГАЦИЯ

Dashboard

Обучение

АНАЛИЗ

Эксперименты

Пакетная обработка

АККАУНТ

Настройки

Выйти

Dashboard



7

Экспериментов



7

Обработано



1510

Сигналов

Загрузка данных выстрела

Shot ID (например, 30420)

Загрузить

Сравнение моделей

Модель	Тип	Accuracy	F1	AUC	Статус
transformer_classification	transformer	0.9733	—	0.9415	Активна
Autoregressive Transformer	transformer	0.9330	0.9390	1.0000	Активна
bi-LSTM+Attention (Disruption)	lstм_attention	1.0000	1.0000	1.0000	Активна
Random Forest Baseline	random_forest	0.9733	—	0.9677	Активна

Список загруженных экспериментов (7 выстрелов MAST)

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Batch

Settings

Admin МЕРФИ (admin)

Выйти

bi-LSTM+Attention

lstm_attention

0.9733

—

0.9938

Активна

Загруженные эксперименты

Shot ID	Источник	Статус	Дата	Сигналов	
11770	MAST	preprocessed	06.05.2026, 11:42:35	294	Открыть
11760	MAST	preprocessed	06.05.2026, 11:39:41	275	Открыть
11730	MAST	preprocessed	06.05.2026, 11:37:27	232	Открыть
11720	MAST	preprocessed	06.05.2026, 11:35:21	232	Открыть
11750	MAST	preprocessed	06.05.2026, 10:06:45	243	Открыть
11700	MAST	preprocessed	06.05.2026, 10:04:28	229	Открыть
30420	MAST	preprocessed	05.05.2026, 17:12:24	5	Открыть

- НАВИГАЦИЯ
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- Эксперименты

Пакетная обработка
- АККАУНТ
- Настройки

Выйти

Выстрел #11770

MASTpreprocessed06.05.2026, 11:42:35

Параметры (294)

- Поиск сигналов...
- Все

Сбросить

4 выбрано
- ☒

adg/density_gradient1 / m ** 4
- ☒

adg/gradient_positionm
- ☒

aga/group1electrons/s
- ☒

aga/group2electrons/s
- ☐

aga/group3electrons/s
- ☐

aga/group4electrons/s
- ☐

aga/group5electrons/s
- ☐

aga/group_error0/1
- ☐

aga/helioselectrons/s
- ☐

amb/fl_p4l_1Wb
- ☐

amb/fl_p4l_4Wb
- ☐

amb/fl_p4u_4Wb
- ☐

amb/fl_p5l_1Wb
- ☐

aga/inboard_lowerelectrons/s



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АККАУНТ

Настройки

Выйти

Параметры (294)

Поиск сигналов...

Все Сбросить 4 выбрано

- ☒ adg/density_gradient 1 / m ** 4
- ☒ adg/gradient_position m
- ☒ aga/group1 electrons/s
- ☒ aga/group2 electrons/s
- ☐ aga/group3 electrons/s
- ☐ aga/group4 electrons/s
- ☐ aga/group5 electrons/s
- ☐ aga/group_error 0/1
- ☐ aga/helios electrons/s
- ☐ amb/fl_p4l_1 Wb
- ☐ amb/fl_p4l_4 Wb
- ☐ amb/fl_p4u_4 Wb
- ☐ amb/fl_p5l_1 Wb
- ☐ aga/inboard_lower electrons/s

ML-анализ

Классифицировать

Предсказать срыв

Классификация: Нестабильный

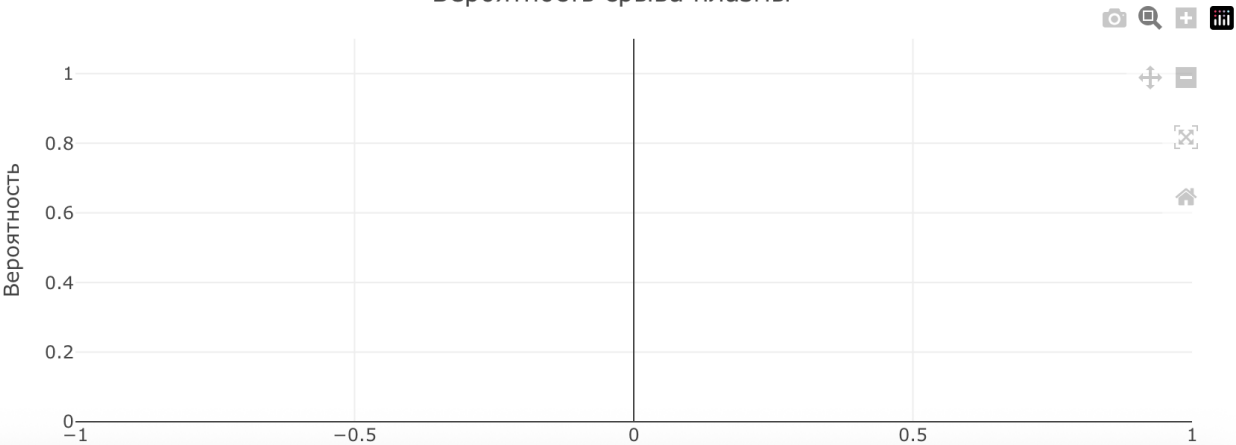
Уверенность: 52.6%

Время инференса: 10549.1 мс

Предсказание срыва: Норма

Время инференса: 12224.6 мс

Вероятность срыва плазмы



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Обучение моделей

Тип модели

Random Forest (baseline)

Ансамбль решающих деревьев. Быстрое обучение, хорошая интерпретируемость. Подходит для базового анализа и сравнения.

bi-LSTM + Attention

Двунаправленная LSTM с механизмом внимания. Хорошо работает с временными рядами, улавливает долгосрочные зависимости.

Transformer

Архитектура на основе self-attention. Лучшая производительность на больших данных, параллельное обучение.

Задача

☒ Классификация
Определение типа плазменного режима

☐ Предсказание срывов
Раннее предупреждение о дестабилизации плазмы

Гиперпараметры (4 параметров)



Набор данных

Shot IDs

Например: 30420, 30425, 30430-30450

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Настройки

Выйти

Начать обучение

История запусков

Обновить

ID	Статус	Прогресс	Метрики	Начало	Завершение
8	Завершено	100.0%	accuracy: 0.9733 f1: 0.9733 auc_roc: 0.9415 epochs: 43.0000	07.05.2026, 13:01:57	07.05.2026, 13:07:28
7	Завершено	100.0%	accuracy: 0.9733 f1: 0.9733 auc_roc: 0.9938 epochs: 22.0000	07.05.2026, 13:01:57	07.05.2026, 13:07:28
6	Завершено	100.0%	accuracy: 0.9733 f1: 0.9733 auc_roc: 0.9677	07.05.2026, 13:01:57	07.05.2026, 13:01:58
5	Ошибка	0.0%	—	06.05.2026, 11:23:12	06.05.2026, 11:23:12
4	Ошибка	0.0%	—	06.05.2026, 10:16:36	06.05.2026, 10:16:37
3	Ошибка	0.0%	—	06.05.2026, 08:54:58	06.05.2026, 08:54:58
2	Ошибка	0.0%	—	06.05.2026, 08:54:58	06.05.2026, 08:54:58

←

→

↺

🏠

https://tokamak-ai.ru/experiment/38

🔑

☆

on

T

📄

Завершить обновление

⋮

⚙️ Tokamak Analysis

Dashboard

⚙️ Training

≡ Batch

⚙️ Settings

Admin MEPhI (admin)

Выйти

Dashboard

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⚡ Эксперименты

≡ Пакетная обработка

АККАУНТ

⚙️ Настройки

→ Выйти

Параметры (294)

Поиск сигналов...

Все

Сбросить

4 выбрано

☒ adg/density_gradient

1 / m ** 4

☒ adg/gradient_position

m

☒ aga/group1

electrons/s

☒ aga/group2

electrons/s

☐ aga/group3

electrons/s

☐ aga/group4

electrons/s

☐ aga/group5

electrons/s

☐ aga/group_error

0/1

☐ aga/helios

electrons/s

☐ amb/fl_p4l_1

Wb

☐ amb/fl_p4l_4

Wb

☐ amb/fl_p4u_4

Wb

☐ amb/fl_p5l_1

Wb

☐ aga/inboard_lower

electrons/s

Предсказание срыва: Норма

Время инференса: 12361.5 мс

Вероятность срыва плазмы

📷

🔍

+

📄

1

0.8

0.6

0.4

0.2

0

—1

—0.5

0

0.5

1

Вероятность

Время (с)

Зона предупреждения

Вероятность срыва

Порог (0.7)

Экспорт

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НАВИГАЦИЯ

 Dashboard

 Обучение

АНАЛИЗ

 Эксперименты

 Пакетная обработка

АККАУНТ

 Настройки

→ Выйти

Настройки

Порог предсказания срывов

Порог вероятности, при превышении которого выдаётся предупреждение о возможном срыве плазмы.

Порог: 0.70



0.0 (чувствительный)

1.0 (строгий)

Уведомления

☒ Включить уведомления

Получать уведомления о завершении обучения и результатах анализа

Сохранить настройки

Аккаунт

Экспорт данных

Удалить аккаунт